Abdur Rahaman

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Indore, Madhya Pradesh, India

Research Interests

- My research focuses on building **efficient, robust, and interpretable machine-learning systems** for biomedical, time-series, and multimodal data. I work on randomized neural networks and Broad Learning Systems, with applications to Alzheimer's disease diagnosis from MRI and EEG, probabilistic time-series forecasting, medical signal analysis, and audio-visual speech enhancement. I am especially interested in methods that remain reliable under limited, noisy, imbalanced, or heterogeneous data.

Education

Indian Institute of Technology (IIT) Indore — *Jul 2024 – present*
Ph.D. in Machine Learning, Department of Mathematics (OPTIMAL Research Lab) — Indore, India

Jamia Millia Islamia (JMI) — *Aug 2020 – Aug 2022*
Master of Science, Department of Mathematics — New Delhi, India

Gauhati University — *Aug 2017 – Jul 2020*
Bachelor of Science in Mathematics — Assam, India

Assam Higher Secondary Education Council (AHSEC) — *2017*
Higher Secondary Examination — Assam, India

Secondary Education Board of Assam (SEBA) — *2015*
High School Examination — Assam, India

Research Experience

Indian Institute of Technology Indore [🌐] — *Jul 2024 – present*
Doctoral Research Scholar (Advisor: Prof. M. Tanveer) — Indore, India

- Develop **robust and computationally efficient randomized neural networks** and Broad Learning Systems for classification under noise, uncertainty, class imbalance, and heterogeneous data.
- Investigate **biomedical AI for Alzheimer's disease**, including MRI-based complex-valued learning and EEG-based graph–state-space generative modeling for diagnosis and data augmentation.
- Work on **multivariate time-series forecasting**, dependence modeling, and multimodal learning, including audio-visual speech enhancement.
- **Outcomes:** First-author and co-authored papers accepted at **WCCI/IJCNN 2026**, **ICONIP 2026**, and MICCAI 2026 satellite workshops; selected for the **IJCNN 2026 Doctoral Consortium** with an INNS travel grant.

Selected Publications

J=Journal, C=Conference, M=Manuscript

[M.1] M. Sajid, A. Quadir, **Abdur Rahaman**, P. N. Suganthan, M. Tanveer. *Uncertainty-Aware Ensemble Deep Randomized Neural Networks for Classification.* arXiv, 2026. [Paper]

[M.2] **Abdur Rahaman**, M. Sajid, M. Tanveer. *CoreX-RVFL: A Unified Framework for Stable, Balanced and Structured Randomized Learning.* Pattern Recognition, 2026 (Revision submitted).

[M.3] **Abdur Rahaman**, M. Sajid, M. Tanveer. *WARM-RVFL: Weight Attention using Relational Metrics Based Randomized Neural Networks.* Neural Networks, 2026 (Under review).

[M.4] **Abdur Rahaman**, M. Tanveer. *Chaos-Randomized Neural Network.* Pattern Recognition, 2026 (Under review).

[C.1] M. Sajid, **Abdur Rahaman**, M. Tanveer. *Beyond Real-Valued MRI Features: Complex Broad Learning for Early Alzheimer's Disease Diagnosis.* **MICCAI MIRASOL Workshop**, 2026. [Paper]

[C.2] M. Tanveer, A. S. Rana, S. Jain, A. Kumar, A. Tiwari, **Abdur Rahaman**, A. Quadir, M. Sajid. *GraM-Diff: A Unified Graph–Mamba Diffusion Framework for EEG-Based Alzheimer's Disease Data Generation and Diagnosis.* **MICCAI EMERGE Workshop**, 2026 (Oral). [Paper] [Code]

[C.3] **Abdur Rahaman**, A. Quadir, M. Tanveer. *Correntropy-Based Robust Learning.* **ICONIP**, 2026 (Accepted). [Code]

[C.4] **Abdur Rahaman**, A. Quadir, M. Sajid, M. Akhtar, M. Tanveer. *ECA-BLS: An Efficient Complex-Augmented Broad Learning System.* **IEEE WCCI/IJCNN**, 2026. [Paper] [Code]

[C.5] **Abdur Rahaman**, A. Quadir, M. Tanveer. *RoBell-RVFL: A Robust Generalized Bell Random Vector Functional Link Network.* **IEEE WCCI/IJCNN**, 2026. [Paper] [Code]

[C.6] A. Quadir, **Abdur Rahaman**, P. N. Suganthan, M. Tanveer. *GBFRVFL: Granular-Ball Computing-Based Fuzzy Random Vector Functional Link Network.* **IEEE WCCI/IJCNN**, 2026. [Paper] [Code]

[C.7] A. Quadir, **Abdur Rahaman**, M. Akhtar, M. Tanveer. *Robust Dual-Model Collaborative Random Vector Functional Link Network.* **IEEE WCCI/IJCNN**, 2026. [Paper] [Code]

[C.8] M. Akhtar, A. Varshney, A. Quadir, **Abdur Rahaman**, M. Arshad, M. Tanveer. *Robust Broad Learning System with Wave Loss for Classification under Data Uncertainty.* **IEEE WCCI/IJCNN**, 2026. [Paper] [Code]

[C.9] M. Sajid, D. Gupta, Y. B. Modi, S. Jain, H. J. S. Ganji, **Abdur Rahaman**, H. Choudhary, N. Saleem, A. Hussain, M. Tanveer. *AUREXA-SE: Audio-Visual Unified Representation Exchange Architecture with Cross-Attention and Squeezeformer for Speech Enhancement.* **INTERSPEECH-W / AVSEC**, 2025. [Paper] [Code]

Honors and Awards

- **Selected for the IJCNN 2026 Doctoral Consortium** by the International Neural Network Society (INNS), with an INNS travel grant — 2026
- **CSIR-UGC NET (Mathematical Science)** – All India Rank 52 — 2023
- **GATE (Mathematics)** – All India Rank 347 — 2024

Academic Services

- **Leadership**
  - Vice Chair, IEEE Student Branch, IIT Indore – Apr 2026–present
  - Student Representative, IEEE Madhya Pradesh Section – 2026–present
  - Webmaster, IEEE CIS Madhya Pradesh Section – 2025–2026
- **Schools, Workshops & Outreach**
  - Co-organizer, IEEE CIS Summer School on Emerging Trends in Computational Intelligence, Deep Learning, and Large Language Models, IIT Indore, Jul 23–25, 2025.
  - Coordinator, IEEE CIS Winter School on Computational Intelligence and Generative AI, IIT Indore, Jan 8–9, 2025.
  - Organizer, IEEE CIS Workshop on Innovation & Leadership in Artificial Intelligence, IIT Indore, Oct 16, 2024.
  - Organizer, IEEE Day Workshop on Membership Benefits of IEEE and IEEE Societies, IIT Indore, Oct 9, 2024.
  - Lead, IEEE CIS High School Outreach Program on foundations of machine learning and AI and their impact on society, Sep 27, 2025.

Technical Skills

- **Programming:** Python, MATLAB, C, R, HTML, CSS, Java, DBMS.
- **ML / Scientific Computing:** PyTorch, TensorFlow, Keras, scikit-learn, Pandas, Matplotlib, OpenCV.

Key Courses

- **Computer Vision**, Jan 2025 - May 2025
- **Theory of Computation and Graph Theory**, Jan 2025 - May 2025
- **Machine Learning**, July 2024 - Dec 2024
- **Algebric Graph Theory**, July 2024 - Dec 2024
- **Probability and Statistical Methods**, July 2024 - Dec 2024